

Cryptocurrency and Associated Bourse Analysis using Machine Learning and Knowledge Graphs

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Abstract— Data science has a goal to discover hidden patterns in raw data. This is exceptionally useful when discussing the stock market. The way crypto currency has taken over the world is something that wasn't predicted 20 years ago. It seems that regular market indicators aren't being implemented in the case of crypto currencies such as bit coin. Crypto currency seems to evade inflation as well. For analysts to perform analyses on these entities there are umpteen tools. One instance is the knowledge graph, which is a collection of interlinked descriptions of objects, events, or concepts. It puts data in context through linking and semantic metadata. Historical data present across many trading and finance websites enable analysts and enthusiasts to try machine learning and deep learning algorithms to predict the subsequent nature of that market. The paper has summarized and analyzed all the possible algorithms to prepare the data available regarding the lifetime and volatility of a currency state. A knowledge graph linking all the related information was also materialized along with a UML-style ontological graph is done to aid and support proper inference from the knowledge representation model.

Keywords— Cryptocurrency, stock, machine learning, deep learning algorithms, data science

I. INTRODUCTION

Cryptocurrency is everywhere. The sudden rise and hike sparked debates discussing its potential capacity to overthrow paper currency. The one thing that sets this apart from the usual currency is that cryptocurrency is fully digitalized [1]. Cryptocurrency also uses a unique technology called Blockchain that has the sole monopoly over the technology of crypto transactions by possessing a digital ledger that records and processes mutual money exchange across customers/buyers. Several analysts are of the opinion that cryptocurrency can have the capacity to defeat inflation. The interest in a common exchange provision with a decentralized banking system surely must have sprung up from the scientific community which led to its founding. It is not affected by any currency rate or government policies. For investors to get a better idea of the efficiency of the currency, predicting future prices has been a constant subject of discussion among analysts who resort to machine learning techniques to fulfill the same [2] The need to link attributes post-analysis for visual representation within the same data set so that it can be represented as a visual tool as an additional support. Knowledge Graphs also allow enhancement of data by including the context of data by knowledge specific to its domain via designated taxonomies, libraries, vocabularies,

and ontologies. By deriving resources from accessible data, a semantic layer of data by making a Knowledge Graph gives data analysts and entrepreneurs a free hand to gain knowledge from data assets. The takeaway would mean an easy-to-understand representation of several complex paradigms that would enlarge the semantic description of information context would enable users to access a machine-readable representation of complex systems that involve a cross-over to a real-time model from a theoretical, knowledge domain.

The concept of a knowledge graph appeared in 2012 when Google essentially changed its string search method to a network/layers of search results or Knowledge graphs that explains the "associated search results"/ Recommended tab as the algorithm learns to link each and every search result by the users. As more avenues arise on a particular topical sphere, the graph keeps on increasing infinitely. We have to keep adjusting and adapting to the fast-paced environment that we live in currently where maybe in a few years, data will be as precious as petroleum.

II. LITERATURE REVIEW

The paper aims to conduct a study by analyzing several instances and datasets of bourses and cryptocurrency mining schemes using both machine learning and knowledge graphs supported by its respective ontology libraries. Here, the authors would be implementing research in their area of interest via knowledge graphs. This will help to learn and understand the relation between different entities that will play a major role in stock markets and cryptocurrency initiatives across the world. Predicting the accurate stock price has been the aim of investors ever since the beginning of the stock market. In order to profit from a cryptocurrency venture, investors must make sure they are making the right choices. However, it is difficult for investors to analyze and forecast the market by churning all this information. This calls for developing a network of data entities that are linked together through some attributes that enable us to get a better understanding of that data set. The idea of knowledge graph goes back in time: in artificial intelligence there was some work in the 1960s, 1970s, 1980s, and 1990s under titles such as semantic memory, semantic networks, knowledge-based systems, knowledge bases, semantic web, and so forth. Companies host millions of linked open data (semantic) networks for participation in social networks. Existing tech tycoons have already manufactured their knowledge graphs

according to the framework of their company and the possible reasons why people would use their services.

To build a knowledge graph, we would need to maintain a knowledge-based management system, that performs rule-based tasks breaking down huge bundles of data and providing tools for data analytics and machine learning by using complex techniques and algorithms [5][6] realizes that knowledge graphs have emerged as a standard for structured knowledge representation, whereby entities and their inter-relations are represented and conveniently stored as (subject, predicate, object) triples in a graph that can be used to power various downstream applications. Since these tools rely on creating linkages via ontology libraries, some papers talk about the role played by these libraries in designing one. These libraries allow structured definitions of data that open up new possibilities, for instance – it eliminates keyword matching by introducing intelligent searching tools and document exchange through ontology mappings or documents [7]. For the ontology-based linking system, the system used here will mostly derive from OWL and RDF- related libraries while creating links for the knowledge graphs. The concept of ontology can be applied in constructing information retrieval interfaces through an ontology-based web information retrieval system.

Through this approach, the library functions as an active information retrieval system that steers the attention of end-users through an interface (primarily ontology-driven) that is set for search criteria definition. To start with, a user first selects a relevant domain in order to start building a query. The interface then provides several search points along with their descriptions. The web information elements are rolled back when the user selects the required ontology entities by selecting the appropriate ontology-to-web links. It must be kept in mind that while building an ontology library, the field of application, as well as its feasibility to the selected area of study by the researcher or the stakeholder, must be kept in mind [9]. Different ontology libraries have to be used for different purposes. Examples of some numerous ontology languages that were developed in the last few years include Resource Description Framework (RDF) and RDF Schema, and the Ontology Inference Layer (DAML + OIL) [11].

III. MATERIALS, METHODS AND DISCUSSION

To pursue the study, the authors chose three datasets, which are described below. It should be noted that the name 'Bourses' was chosen to consider non-American markets as well.

A. Key Terminologies

- Date – Date of stock recording
- Open, Close: The starting and closing price of the stock at the start and end of the day
- High and Low: Price Peak of the stock and lowest price of the stock that day.
- Volume: The trading volume of a stock that denotes the number of stocks being bought that day. (**Target Variable**)

- Closest Bollinger Band (CBB): An indicator that is used to define the volatility of the stock; this was put in as a target label to support supervised learning-related algorithms[12]. Days with Close prices below the Moving Average are put under a lower Bollinger Band and vice-versa.

B. Description Of Datasets

- Ford dataset – Table I contains a snippet of the six year historical price dataset with a total of 1638 rows and 7 columns.

TABLE I. FORD DATASET

Date	Open	High	Low	Close	Volume	CBB
2015-01-30	14.75	14.89	14.63	14.71	304397	Low
2015-02-02	14.75	15.3	14.74	15.27	403718	High
2015-02-03	15.47	15.79	15.44	15.65	483354	High

- Bitcoin dataset: Snippet of historical prices of the world's most famous cryptocurrency that has a total of 7 columns and 2750 rows in Table II.

TABLE II. BITCOIN DATASET

Date	High	Low	Open	Close	Volume	CBB
2013-01-01	777.51	713.59	763.28	735.07	468627	Low
2013-01-08	834.15	789.11	714.74	818.71	403718	High
2013-01-09	859.51	801.66	815.44	859.51	483354	Low

- Cryptodata : A snippet of the dataset to design a Knowledge Graph, as described in Table III that includes both text and numeric fields.

TABLE III. CRYPTODATA DATASET

Name	Prices				Volume	Country
	Open	Close	High	Low		
Aave	53.4	40.73	53.29	42.40	58309	Switzerland
Binance G	0.10	0.099	0.105	0.105	20093	Cayman Islands
Binance P	0.10	0.100	0.105	0.107	34449	Seychelles

C. Algorithms Used

- *Classification algorithm* that acts as a boosting algorithm: LightGBM which is a type of Gradient Boosting Algorithm.
- *Time Series Forecasting Algorithm*: The main aspect of predicting the stock-related data is its variance with time. We can project the possible price of the dataset

when it reaches a specific time. We chose two algorithms namely Prophet and ARIMA.

- *Artificial Neural Networks:* The process starts with the input layer being fed with the required input. Linear transformations are performed by the neurons with the corresponding weights and biases. The formula to deduce an activation function is described as –

$$Y = \text{Activation}(Z(\text{input} * \text{weights}) + \text{bias}) \quad (1)$$

- *Knowledge Graphs:* For a detailed description of the process, kindly refer to Experiments.

IV. EXPERIMENTS

A. Classification Algorithms

Boosting algorithms include generating and detecting weak learners and then combining all of them to form one strong entity. To increase the accuracy of the data set, they are joined together to form a stronger learning attribute [3][14]

1) LightGBM: What, Why, and How?

LightGBM is a gradient boosting algorithm that is based on decision tree learning algorithms. For this dataset, there was some surface analysis done to fine-tune the parameters given in Table IV with the train-test ratio as 80-20 and hence generate a related heat map with the confusion matrix represented here in Fig.1.

TABLE IV. LIGHTGBM PARAMETERS

Name of Parameter	Functions/Uses
make_classification()	Generate a random n-class classification problem
n_informative	No of informative features around a hypercube dataset
n_redundant	Generated as random linear combinations, this is the count of informative yet redundant features present in the dataset.

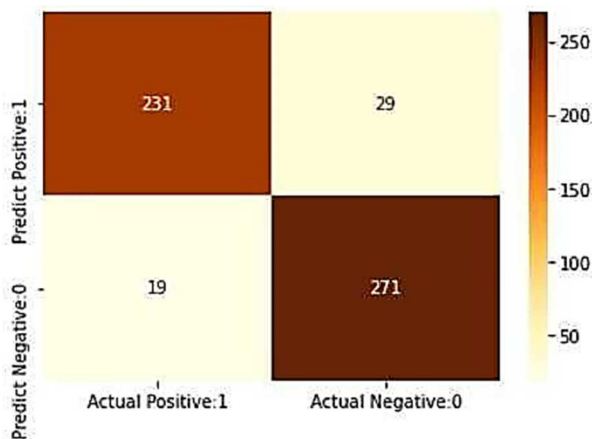


Fig. 1. LightGBM Confusion Matrix

The accuracy of the model was deduced to be 91% for 'gbdt' mode. For LightGBM, three main methods are very much in practice which includes GOSS, DART, and GBDT (used here), whose model accuracies were plotted together in a box and whisker plot as shown in Fig.2. In short, LightGBM can deal with huge datasets consuming less memory using parallel and GPU learning. This accounts for its accuracy, fast training mechanism, and high efficiency. These factors are what sets this model apart from the other models.

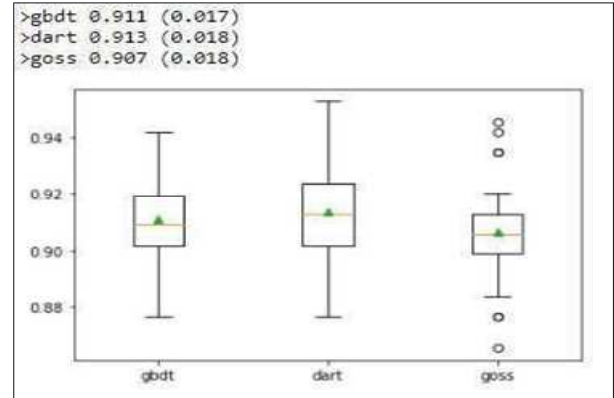


Fig. 2. Box Whisker Plot for accuracy

B. Time Forecasting Algorithms

1) FBProphet

Developed by Facebook's Core Data Science Team, FBProphet is widely used in machine learning for forecasting time series for instances that involve time series data with all kinds of seasonalities (yearly, weekly and monthly) including holidays and vacations [15].

Prophet is also suitable for historical data with several seasons. To carry out the process of regression, FBProphet uses time as a regression variable (regressor) along with the time series' linear and non-linear parameters as components. The data can be fitted into the model which can be changed from linear (default) to non-linear in FBProphet as per the requirements. Table V has the important parameters used for the same.

TABLE V. FBPROPHET PARAMETERS

Parameter	Functions/ Uses
Prophet()	Performing forecasting
n_changepoints	Setting the number of potential changepoints
yearly_seasonality	Fit yearly seasonality.

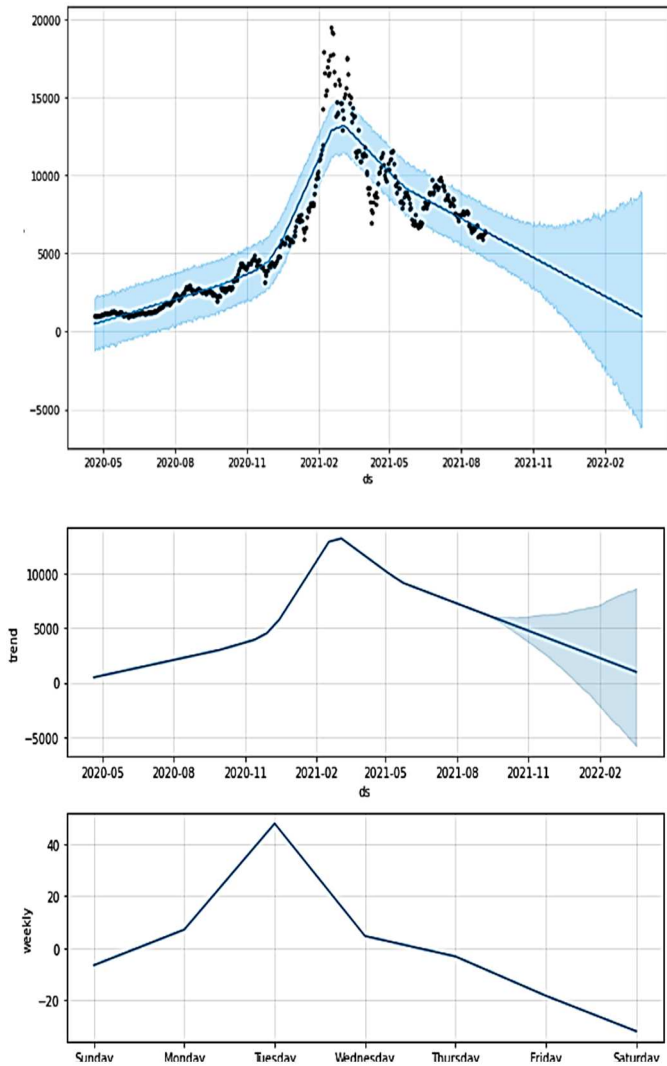


Fig. 4. Fitting the data and forecasting data (in blue) in graph with seasonality.

2) ARIMA

ARIMA (AutoRegressive Integrated Moving Average) is an autoregressive statistical model used to forecast a time series in the future using the series past values using three main steps. Parameters are usually as described in Table VI but may change as per user and situational requirements.

- Identification: The stationary nature of the data series is an indicator for the inclusion of the seasonality factor during implementation.
- Estimation: Finding out constraints such as least square errors as well as maximum likelihood estimation.
- Validation: Testing the validity of the model by pitting the results with the assumptions to reach optimum.

TABLE VI. ARIMA PARAMETERS

Parameter	Functions/Uses
adfuller()	A most common test to indicate the stationary nature of a Time Series that is set to be tested.

test	Type of unit root test to use to detect stationarity if stationary is False and d is None.
rcParams.update.plot_predict	Plot the predicted graph
start_P	P represents the order of the differential of the auto- regressive portion of the ARIMA. start_P denotes its initial endpoint.

During the process of forecasting for future values of a given dataset, it is helpful to assume that each and every point is independent of one another for a given time-series. For this, we may need to retain the stationary nature of the past instances. Most properties of a statistical system don't change over time for a data that is stationary. It should be stationary in order to use ARMA(p, q) (a short way of saying ARIMA(p, 0, q)). The I in ARIMA defines itself as an indicator to identify whether the data is stationary or seasonal to forecast future values. Though ARIMA can process both stationary and seasonal values, stationary data is mostly preferred. Fig.5 plots the graph with both usual and seasonal differencing where seasonal refers a longer period/interval forecasting.

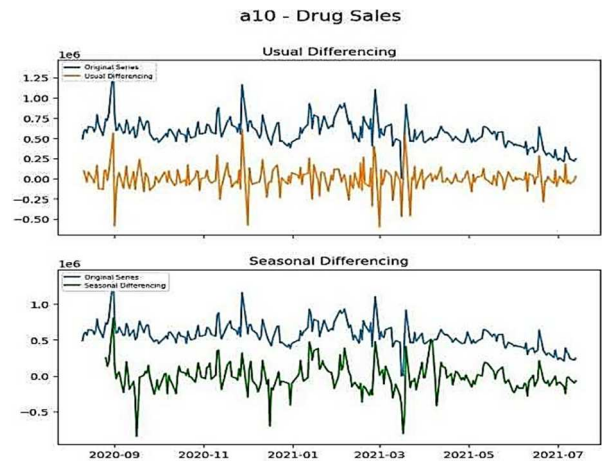


Fig. 5. Usual and seasonal differencing plots

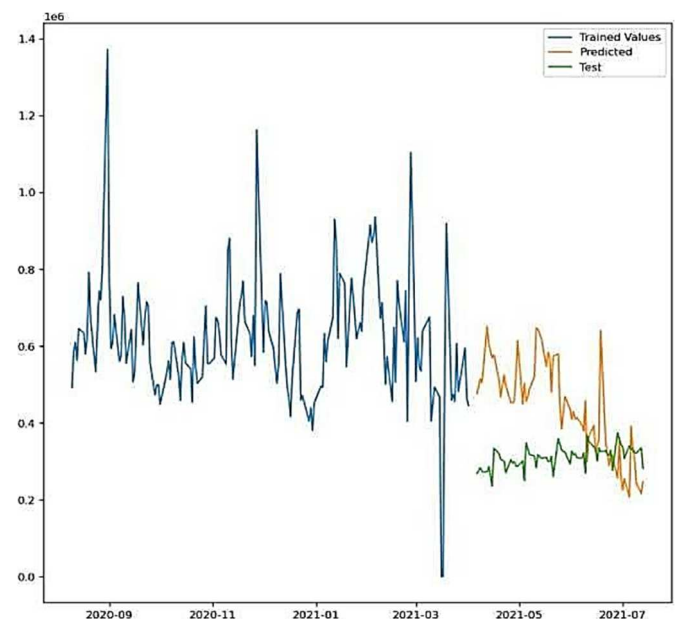


Fig. 6. Trained value, predicted value(orange), and test value plots

C. Neural Networks

The idea of neural networks began in the 1950s-1960s around the concept of an artificial neuron which is called a perceptron [4], which produces a single output having given multiple inputs. If the sum of assigned weights is less than the threshold value, it is 1.

$$\begin{aligned} w_1x_1 + w_2x_2 + \dots w_nx_n &> \theta \\ w_1x_1 + w_2x_2 + \dots w_nx_n &\leq \theta \end{aligned} \quad (2)$$

Else if it is more than the threshold/ target function or equation we can assign it a 0, where θ is a threshold value. Neural networks can learn from earlier instances. Some input from the data is combined with the weights that it assigns to that particular label, whose signals “activate” the neuron after passage through an activation function.

A single node, as shown in Fig.7, belonging to a neural network may turn on or off when there is input through the network. Each layer's output layer becomes the subsequent layer's input, which commences by inputting the data into the first input layer. A neural network is a complete structure with indefinite input, output and hidden layers like in Fig.8. Consider any artificial neuron with function,

$f_j(\mathbf{x}) = (\mathbf{x}_1, \dots, \mathbf{x}_d)$ with some connection weighted vectors

$\mathbf{W}_{jd} = (\mathbf{W}_{j,1}, \dots, \mathbf{W}_{j,d})$, completed by a neuron bias b and associated to an activation function f .

The activation function, in any neural network always plays a key role in taking one vital decision; the decision to activate the neuron, or go for other alternatives. This is also, as mentioned earlier referred to as threshold or transformation to converge the neural networks. Beyond normalizing the output, such functions aid in backpropagation, along with updating the loss function, using the gradient descent curves to reach their minima.

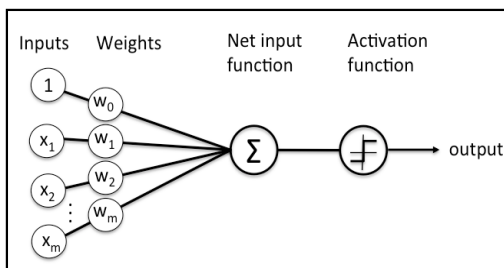


Fig. 7. Representation of a single node. Downloaded from <https://link.springer.com/article/10.1007/s13157-019-01165-8/figures/10>

Some activation functions are mentioned below –

- Identity

$$f(x) = x \quad (3)$$

- Sigmoid/Logistic

$$f(x) = \frac{1}{1+e^{(-x)}} \quad (4)$$

- Hyperbolic Tangent/Tanh

$$f(x) = \frac{2}{1+e^{-x}} - 1 \quad (5)$$

- Hard Threshold

$$\phi \cdot \beta = 1 \quad x \geq \beta \quad (6)$$

- Rectified Linear Unit/ReLU

$$f(x) = \max(0, x) \quad (7)$$

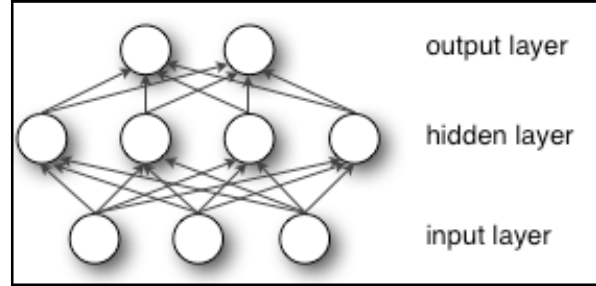


Fig. 8. Sketch of a neural network structure. Image downloaded from <https://www.researchgate.net/publication/335090288>.

Of the three important types of neural networks that are widely used, namely ANN (Artificial Neural Networks), CNN (Convolutional Neural Networks), and RNN (Recurrent Neural Network), this paper will implement Artificial Neural Networks (ANN).

I. Using artificial neural networks for machine learning:

The use of ANNs in the field of deep learning mechanisms has become increasingly popular in applications such as industries and business environments that need model optimization. ANN is a useful neural network mechanism that mimics the biological nerve endings of the human brain [10].

II. Regression with artificial neural networks: ANNs can be used for both classification and regression as well. In situations where financial data involves a lot of statistics, it is ideal that we go for algorithms such as regression that are suited for financial forecasting. For this data set, both Python3 and RStudio with Keras and TensorFlow library have been used to materialize the same.

$$\text{Input Layer} = n_{\text{samples}} + 1 \quad (8)$$

$$\text{Hidden Layer} = \frac{n_{\text{samples}}}{1 + \sum(n_{\text{neurons}} * \text{out}_{\text{neurons}})} \quad (9)$$

To start with, a neural network was generated in RStudio using the current dataset displayed in Fig.9. Input and hidden layers within the dataset were deduced using the total number of samples and attributes that were trained. For the output layer, both linear and ReLU activation functions were used. Calculation of Root Mean Squared Error (MSE) and Mean Absolute Error (MAE) was carried out with a train-test ratio of 80-20 with epochs set to 30 and batch size set to 150. Another activation function called Swish, developed under the Google Residency Programme, was implemented to compare its efficiency with the most popular one, ReLU (Rectifier Linear Unit).

$$f(x) = x * \text{sigmoid}(x) \quad (10)$$

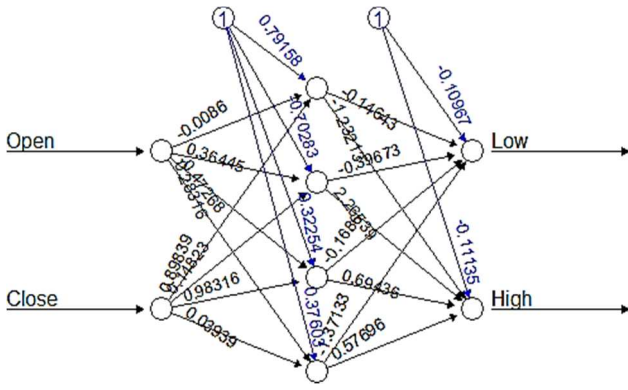


Fig. 9. Neural Network for Bitcoin dataset. Image generated using R and RStudio (v 3.0.1)

Neural networks are usually implemented using the parameters in Table VII.

TABLE VII. ANN REGRESSION PARAMETERS

Parameter	Description
Sequential()	Good for layers in between an input and an output layer tensor. Increase in the frequency of the function call, there is an increase in the number of layers.
kernel_initializer	Assigns weights to Keras layers.
batch_size	Training examples for each forward and/or backward pass.
epoch	A set of all forward and backward passes for all attrib
verbose	Progress bar after epoch completion

Going further, by making the second set of neural networks by using the swish activation function to look for changes in errors.

$$f(x) = \frac{x}{1+e^{-x}} \quad (11)$$

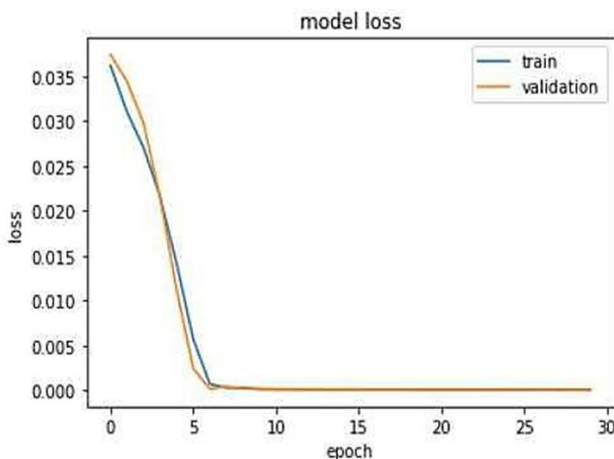


Fig. 10. Loss graphs for ReLU

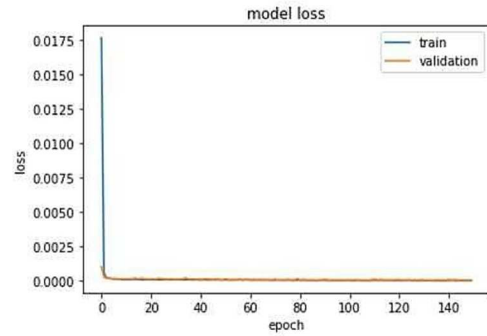


Fig. 11. Loss graphs for Swish

D. Knowledge Graph Visualization

Of the many tools available for semantics analysis, the knowledge graph is an important topic. KGs aid in finance analysis, semantic development, customer sentiment analysis, speech recognition, and much more. A common trait found is that the pre-existing instances contain a lot of sentiment collected via news [9][11], which ignores the semantic features of structured events. RDF (Resource Development Framework) is an instance of an ontology library. This combined with other libraries like OWL and FOAF can take up the metadata present in the data set to create links.

Vocabulary descriptions (schemas, ontologies) written in the RDF Schema language are legal RDF graphs. OpenRefine was used to make this possible. RDF is mainly available in three forms: N3/Notation3- which is the detailed form of the framework, Turtle- which is a simplified RDF- only subset of N3 and N-Triples which emphasizes its other attributes in the form of a subject, predicate, and object. This is explained below.

Aave priced at \$XX is based in Switzerland.
 (subject) (predicate) (object)

Here is a snippet of the linkages created using RDF and OpenRefine. This is to be mapped to the first and second fields of the Cryptodata Table.

```
<rdf:Description>
<rdf:about="http://127.0.0.1:3333/0">
<rdf:type>High</rdf:type>

<rdf:object>Low</rdf:object><rdf:object>Open</rdf:object><rdf:object>Close
</rdf:object></rdf:Description>

<rdf:Description>
<rdf:about="http://127.0.0.1:3333/1"
  <rdf:object>53.29196931</rdf:object><rdf:
  type>53.40227002</rdf:type><rdf:object>40
  .73457791</rdf:object>
<rdf:object>42.40159861</rdf:object><owl:ta
  rgetIndividual>583091.4597628</owl:targetIn
  dividual><owl:members></rdf:Description>
<rdf:Description>rdf:about="http://127.0.0.
  1:3333/0">

<rdf:type>High</rdf:type>

<rdf:object>Low</rdf:object>
```

VI. CONCLUSION

The popularity of cryptocurrencies had skyrocketed in 2017 due to several consecutive months of super-exponential growth of their market capitalization. Research shows that the brain likes to work based on association and has a tendency to connect every idea or skill that it encounters in its lifetime, which is enabled by the mind-map-like the KGs [16]. Such visualization tools involve a unique combination of imagery, color, and visual-spatial arrangement which is proven to significantly improve recall when compared to conventional methods of documentation.

This leads to individuals of different learning speeds and abilities to understand the links between financial entities involved for any instance, which in this case might be termed as the need to predict the price of digital currency. Knowledge graphs are usually used to derive sentiments about a particular entity from financial news and then draw links and then visualize it accordingly [17][18]. A knowledge graph linking all the related information was also made along with a UML-style ontological graph that was done to support the RDF, RDFS, and OWL ontologies used.

Conflicts of Interest

The authors declare that they have no mutual conflicts regarding the subject of research.

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```
<rdf:object>Open</rdf:object>
<rdf:object>Close</rdf:object>
</rdf:Description>
<rdf:Description
  rdf:about="http://127.0.0.1:3333/1">
<rdf:object>0.10901</rdf:object>
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</owl:targetIndividual>
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</rdf:Description>
```

V. RESULTS

The confusion matrix generated for the Bitcoin dataset, calculated its accuracy to 91%. A box and whisker plot shown in Fig.2 was drafted to support this argument. Fig 3 predicts the possible fall in the stock prices of cryptocurrency from the given forecast date up to a certain date which is well explained due to the current pandemic. Fig.4 forecasts the same result this time with seasonality added as a parameter. Fig 5 and 6 show the forecasted graph with ARIMA algorithm plotted with usual and seasonal differencing. Fig 6, in particular shows the predicted graph obtained from splitting the historical data post training and testing.

In addition to this, an artificial neural network was made to act as a regressor for detecting the errors. It was observed that altering the batch size or size of epochs does not result in much improvement of the above metrics.

Losses due to error were found to be comparatively less in Swish drafted in Fig 11, than in ReLU, shown in Fig 10, making it to be the better activation function when it comes to error minimizing. We may look forward to using more of such activation functions in the future. Finally, to conclude the study, a knowledge graph was visualized using the various ontology libraries RDF, RDFS and OWL. The knowledge graph was materialized with several visual tools and was obtained as shown in Fig.12.

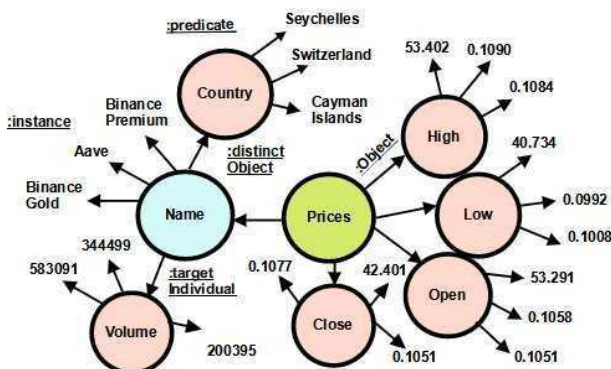


Fig. 12. Knowledge Graph visualization

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